Problem 14.2.9

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \exp(-z^2)$$

Solution

f is entire since f' is defined everywhere, therefore by Cauchy's integral theorem

$$\boxed{\oint_C f(z)\ dz = 0}$$

Problem 14.2.11

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{2z-1}$$

Solution

f' is not defined at $z = \frac{1}{2}$ and therefore Cauchy's integral theorem does not apply. By 14.2.(3), $\oint_C (z-z_0)^{-1} = 2\pi i$. Therefore,

$$\begin{aligned}\oint_C f(z)\ dz &= \frac{1}{2}\oint_C \left(z - \frac{1}{2}\right)^{-1}\ dz \\ &= \boxed{\pi i}\end{aligned}$$

Problem 14.2.13

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{z^4 - 1.1}$$

Solution

f' is not defined at $z = \sqrt[4]{1.1} \approx 1.02$, however this is outside the unit circle and therefore Cauchy's integral theorem applies.

$$\boxed{\oint_C f(z)\ dz = 0}$$

Problem 14.2.18

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{4z-3}$$

Solution

f' is not defined at $z = \frac{3}{4}$ and therefore Cauchy's integral theorem does not apply. By 14.2.(3), $\oint_C (z-z_0)^{-1} = 2\pi i$. Therefore,

$$\begin{aligned}\oint_C f(z)\ dz &= \frac{1}{4}\oint_C \left(z - \frac{3}{4}\right)^{-1}\ dz \\ &= \boxed{\frac{\pi i}{2}}\end{aligned}$$

Problem 14.2.22

Evaluate the integral. Does Cauchy's theorem apply? Show details.

$$\oint_C \operatorname{Re} z\ dz$$

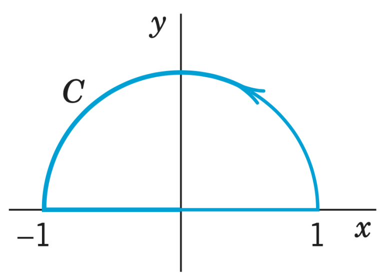

Figure 1: C

Solution

$$f(z) = \operatorname{Re} z = u(x,y) + iv(x,y)$$

$$\begin{aligned} u &= x & v &= 0 \\ u_x &= 1 & v_x &= 0 \\ u_y &= 0 & v_y &= 0 \end{aligned}$$

$u_x \neq v_y$ therefore f is not analytic and Cauchy's theorem does not apply.

$C_1 : z = e^{it}, dz = ie^{it}\ dt,\ \operatorname{Re} z = \cos t, \quad t \in [0, \pi]$

$C_2 : z = t, dz = dt,\ \operatorname{Re} z = t, \quad t \in [-1, 1]$

$$\begin{aligned}
\oint_C f(z)\ dz &= \oint_C f(z(t))\dot{z}(t)\ dt \\
&= \int_0^\pi \cos t i e^{it}\ dt + \int_{-1}^1 t\ dt \\
&= \int_0^\pi \cos t i(\cos t + i\sin t)\ dt + \int_{-1}^1 t\ dt \\
&= i\int_0^\pi \cos^2 t\ dt - \int_0^\pi \cos t\sin t\ dt + \int_{-1}^1 t\ dt \\
&= \frac{i}{2}\int_0^\pi 1+\cos 2t\ dt - \frac{1}{2}\int_0^\pi \sin 2t\ dt + \int_{-1}^1 t\ dt \\
&= \frac{i}{2}\left[t+\frac{\sin 2t}{2}\right]_0^\pi + \frac{1}{4}\left[\cos 2t\right]_0^\pi + \frac{1}{2}\left[t^2\right]_{-1}^1 \\
&= \frac{i}{2}\left[\pi\right] + \frac{1}{4}\left[0\right] + \frac{1}{2}\left[0\right] \\
&= \boxed{\frac{\pi i}{2}}
\end{aligned}$$

Problem 14.2.24

Evaluate the integral. Does Cauchy's theorem apply? Show details. *(Use partial fraction)*

$$\oint_C \frac{dz}{z^2-1}$$

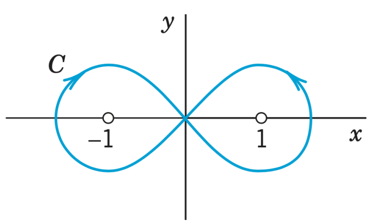

Figure 2: C

Solution

$$\frac{1}{z^2-1} = \frac{A}{z-1} + \frac{B}{z+1} \implies 1 = A(z+1) + B(z-1)$$

$z=-1: \quad B = -\frac{1}{2}$
$z=1: \quad A = \frac{1}{2}$

$$\frac{1}{z^2-1} = \frac{1/2}{z-1} - \frac{1/2}{z+1}$$

The integral becomes,

$$\oint_C \frac{dz}{z^2-1} = \frac{1}{2}\oint_C \frac{1}{z-1} - \frac{1}{z+1}\ dz$$

Splitting C into C_1, the left half that circles $z = -1$ clockwise, and C_2, the right half that circles $z = 1$ counterclockwise, we can evaluate each integral separately.

The first integral has a singularity at $z = 1$, therefore over C_1 Cauchy's theorem holds and the integral is zero. Over C_2, Cauchy's theorem does not hold, but 14.2.(3) gives a solution of πi.

$$\frac{1}{2}\oint_C \frac{1}{z-1} = \frac{1}{2}\left[\oint_{C_1} \frac{1}{z-1}\ dz + \oint_{C_2} \frac{1}{z-1}\ dz\right] = \frac{1}{2}\left[0 + 2\pi i\right] = \pi i$$

The second integral has a singularity at $z = -1$, therefore over C_1 Cauchy's theorem does not hold, but 14.2.(3) gives a solution of $-\pi i$ (due to the clockwise path). Over C_2, Cauchy's theorem holds and the integral is zero.

$$\frac{1}{2}\oint_C \frac{1}{z+1} = \frac{1}{2}\left[\oint_{C_1} \frac{1}{z+1}\ dz + \oint_{C_2} \frac{1}{z+1}\ dz\right] = \frac{1}{2}\left[0 - 2\pi i\right] = -\pi i$$

The final integral is

$$\boxed{\oint_C \frac{dz}{z^2-1} = 2\pi i}$$

Problem 14.3.3

Integrate by Cauchy's formula counterclockwise around the circle.

$$f(z)\frac{z^2}{z^2-1}, \qquad |z+i| = 1.4$$

Solution

$f(z)$ has singularities at

$$z^2 - 1 = 0 \implies z_{1,2} = \pm 1$$

$$|z_{1,2} + i| = \sqrt{2} \approx 1.41 > 1.4$$

Therefore the singularities are outside the domain and Cauchy's theorem holds.

$$\boxed{\oint_C f(z)\ dz = 0}$$

Problem 14.3.11

Integrate counterclockwise.

$$\oint_C \frac{dz}{z^2+4}, \quad C : 4x^2 + (y-2)^2 = 4$$

Solution

$f(z)$ has singularities at

$$z^2 + 4 = 0 \implies z_{1,2} = \pm 2i$$

$z_1 = 2i$ is in the center of the region, however $z_2 = -2i$ lies outside. Therefore we can split the function in to the analytic and non-analytic factors and integrate via Cauchy's integral formula.

$$\begin{aligned}
\oint_C \frac{dz}{z^2+4} &= \oint_C \frac{1}{(z-2i)(z+2i)}\ dz \\
&= \oint_C \frac{1/(z+2i)}{(z-2i)}\ dz \\
&= \frac{1}{(z+2i)}\bigg|_{z_0=2i} \cdot 2\pi i \\
&= \frac{1}{4i} \cdot 2\pi i \\
&= \boxed{\frac{\pi}{2}}
\end{aligned}$$

Problem 14.3.13

Integrate counterclockwise.

$$\oint_C \frac{z+2}{z-2}\ dz, \quad C : |z-1| = 2$$

Solution

Problem 14.3.15

Integrate counterclockwise. *(compute it for two squares: one with vertices $\pm 2, \pm 2i$; the other one with vertices $\pm 4, \pm 4i$)*

$$\oint_C \frac{\cosh\left(z^2 - \pi i\right)}{z - \pi i}\ dz, \quad C \text{ the boundary of the square with vertices } \pm 2, \pm 2, \pm 4i$$

Solution

Problem 14.4.1

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{\sin z}{z^4}\ dz$$

Solution

Problem 14.4.3

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{e^z}{z^n}\,dz, \quad n = 1, 2, \cdots$$

Solution

Problem 14.4.6

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{dz}{(z-2i)^2(z-i/2)^2}$$

Solution

Problem 14.4.11

Integrate. Show the details. *Hint.* Begin by sketching the contour. Why?

$$\oint_C \frac{(1+z)\sin z}{(2z-1)^2}\,dz, \quad C: |z-i| = 2 \text{ counterclockwise}$$

Solution